

OPTIMIZATION OF GLOBAL ONCOLOGY BI THROUGH DATAKU ELT MIGRATION AND DECOUPLING OF DATA TRANSFORMATION LAYERS

INTERNSHIP PROJECT REPORT

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CERTIFICATE

Certified that this Project report titled “**Optimization of Global Oncology BI through Dataiku ELT Migration and Decoupling of Data Transformation Layers**” by “Harshada Deshingkar (Roll No. 31)” is approved by me for submission. Certified further that, to the best of my knowledge, the report represents work carried out by the student as the project as prescribed by the Vishwakarma Institute of Technology Pune in the academic year 2025-26.

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ABSTRACT

This report details the strategic optimization of the Global Oncology Business Intelligence (BI) framework conducted during a professional internship at MSD within the Digital Human Health (DHH) division. The core of the project addressed the architectural challenges inherent in legacy reporting systems, specifically the heavy reliance on Power BI's frontend for complex data processing. By implementing a robust Dataiku ELT pipeline, the project aimed to decouple data transformation layers from the visualization interface. This migration was engineered to eliminate the computational burden of Power Query and DAX-heavy architectures, thereby enhancing long-term scalability, data governance, and reporting performance for critical financial assets such as Keytruda.

The methodology followed a rigorous eight-phase end-to-end engineering lifecycle, beginning with a comprehensive requirement analysis and system discovery to capture the nuances of global financial and market-specific data. This was followed by a technical logic migration, where frontend transformations were translated into upstream Dataiku recipes and orchestrated within an Amazon S3-backed data environment. The development prioritized the creation of high-quality, curated datasets and automated pipelines that allow for step-wise debugging and modular expansion. Rigorous quality control and reconciliation against legacy outputs were performed to ensure absolute financial integrity and consistency across all key performance indicators (KPIs).

The final implementation resulted in a streamlined, analysis-ready data architecture that significantly improved dashboard refresh stability and reduced transformation latency. By centralizing the business logic within a scalable ETL/ELT environment, the project successfully delivered a flexible reporting solution capable of performing complex trend identifications, variance analyses, and forecast-versus-actual evaluations.

CHAPTER 1

INTRODUCTION

I am working as intern at MSD Hyderabad Global Capabilities Center India Private Limited (Kharadi, Pune) during the period 12th January 2026 to 30th June 2026. Wherein, I've worked on the development of financial and performance reporting solutions based on sales data for Keytruda across multiple clinical indications. My responsibilities involved consolidating and validating data from various sources, performing data preprocessing and transformations using Dataiku, and designing interactive dashboards in Microsoft Power BI. A substantial portion of the work required advanced DAX measures to enable robust calculations and flexible reporting views. In addition, I created analytical comparison reports, including forecast-versus-actual evaluations and prior-year (PY) performance comparisons, to support trend identification and variance analysis. These reports were designed to provide stakeholders with accurate, timely insights to support evidence-based and operational decision-making.

1.1 About Organization

MSD (Merck Sharp & Dohme) is a global, research-driven biopharmaceutical company engaged in the discovery, development, manufacturing, and commercialization of innovative medicines and vaccines. The organization operates across the full healthcare value chain, with strong capabilities in research and development, clinical trials, regulatory affairs, quality assurance, and large-scale manufacturing, enabling the delivery of science-based solutions to patients worldwide. MSD's work spans multiple therapeutic areas, including oncology, vaccines, infectious diseases, and immunology, with a strategic emphasis on advancing therapies that address significant unmet medical needs. Through sustained investment in scientific innovation and collaboration with healthcare systems, academic institutions, and research partners, MSD aims to improve health outcomes by enabling access to safe, effective, and evidence-based treatments.

1.2 About Division, DHH

Digital Human Health (DHH) is a key division at MSD that focuses on advancing digital innovation in healthcare through the strategic use of data, analytics, and technology-enabled solutions. DHH supports the organization's broader mission by developing and scaling digital capabilities that enhance operational effectiveness and enable more informed decision-making across the product lifecycle. The function typically collaborates with cross-functional stakeholders to design data-driven frameworks, build analytical products, and implement digital tools that translate complex healthcare data into actionable insights. By prioritizing responsible data stewardship, privacy, and governance, DHH aims to ensure that digital solutions are reliable, compliant, and aligned with patient and public health needs, thereby contributing to more efficient, outcome-focused healthcare delivery.

1.3 Products and Services

MSD's pharmaceuticals portfolio encompasses a wide range of prescription medicines developed to address unmet medical needs across acute and chronic conditions. This work involves end-to-end activities spanning drug discovery, preclinical research, clinical development, regulatory submissions, and post-marketing safety surveillance, supported by robust quality systems and global manufacturing capabilities. Across therapeutic areas, MSD emphasizes evidence-based use of medicines through clinical trial data and real-world evidence generation, enabling healthcare stakeholders to make informed, guideline-aligned treatment decisions.

 **Oncology:** Focus centers on advancing innovative cancer therapies, including approaches designed to improve outcomes across multiple tumor types and stages of disease. Within the global oncology market, **KEYTRUDA (pembrolizumab)** is widely regarded as MSD's **flagship, market-leading product** and one of the **top-selling oncology medicines worldwide**, generating approximately **US\$ 31.7 billion** in annual revenue and accounting for roughly **half of the company's total product revenue**. The company's oncology work integrates translational science, biomarker-driven research, and rigorous clinical development programs to evaluate efficacy, safety, and patient-relevant endpoints such as

survival and quality of life. In addition, MSD supports ongoing evidence generation and lifecycle management to refine appropriate use in clinical practice and to expand understanding of how therapies perform across diverse populations and healthcare settings.

✚ **Vaccines:** MSD's vaccines work is aimed at preventing infectious diseases and reducing population-level morbidity through the development, evaluation, and global supply of immunizations. Within its vaccine portfolio, **GARDASIL/GARDASIL 9**, a vaccine indicated for the prevention of disease caused by certain types of **Human Papillomavirus (HPV)** is widely regarded as one of MSD's flagship vaccines and a major contributor to the company's vaccine franchise. This work includes conducting large-scale clinical studies to establish safety and effectiveness, maintaining stringent quality standards in manufacturing, and supporting pharmacovigilance systems to monitor vaccine safety after deployment. By collaborating with public health stakeholders and healthcare systems, MSD contributes to prevention strategies that can improve long-term health outcomes across different age groups.

✚ **Immunology:** MSD's immunology efforts focus on diseases driven by immune dysregulation, where targeted modulation of immune pathways can reduce symptoms, prevent progression, and improve patient functioning. The organization applies mechanistic research and clinical evidence generation to identify therapeutic targets, evaluate treatment effects, and support appropriate use in clinical care. This work is complemented by safety monitoring and long-term follow-up, recognizing that immune-directed therapies often require careful benefit-risk assessment over extended periods.

✚ **Infectious diseases:** MSD's work in infectious diseases addresses both established and emerging pathogens through the development of antiviral and antimicrobial strategies, as well as prevention-oriented approaches where applicable. The company's efforts often emphasize stewardship, surveillance, and evidence generation to support appropriate use and help mitigate public health challenges such as antimicrobial resistance. Through coordinated research, clinical development, and global access planning, MSD aims to contribute to improved outcomes for infectious disease patients and strengthened preparedness for future outbreaks.

CHAPTER 2

PROJECT CONTEXT AND BACKGROUND

I am part of a data visualization team of over 50 members that supports more than eight regions across the globe. The team gathers reporting requirements from regional stakeholders and develops Power BI reports tailored to their business needs. Work is organized across multiple teams within the IC (Insights Consumption) function, and the reporting deliverables vary based on use case. Key report categories include financial performance reports, healthcare professional (HCP) and healthcare organization (HCO)- based reports, patient analytics reports, and medical representative (MR) / sales representative performance reports.

2.1 Background

The project operates within an internal Business Intelligence (BI) solution that consolidates **global and regional Keytruda sales data** to enable consistent analysis of **revenue trends, key performance indicators (KPIs), and market performance** over time. By integrating datasets across geographies into a unified reporting layer, the solution supports stakeholders in tracking performance and interpreting changes through standardized metrics and visual analytics.

2.2 Reporting Structure (Multi-Level Performance Visibility)

The BI ecosystem provides **multi-level performance visibility** through a tiered reporting structure:

- ✚ **A1: Global overview:** Enterprise-level dashboards and consolidated business summaries (this level remains the primary focus of my team).
- ✚ **A2: Market level:** Country/region-level reporting that supports market-specific tracking and evaluation.
- ✚ **A3: Self-service/ ad hoc support:** Analytical views that test new measures, new tools, business rules, and reporting concepts before broader adoption.

2.3 High-Stakes Financial Reporting Environment

As a flagship product in MSD's oncology portfolio, Keytruda generates large volumes of **financial, sales, and operational data** across multiple regions and reporting cycles. The reporting environment therefore requires strong standards for **accuracy, timeliness, auditability, and scalability**, because dashboards and reports directly inform high-impact business decisions.

2.4 Bottlenecks

The current reporting infrastructure shows several performance and operational constraints:

- ✚ **Refresh latency:** Dataset refresh cycles exceed **45 minutes**, which delays time-sensitive analysis and reduces dashboard responsiveness.
- ✚ **Debugging and QC constraints in Power BI:** Business logic largely resides in **DAX and Power Query**, which limits practical inspection of intermediate tables, filter-based diagnostics, and step-level validation during quality checks.
- ✚ **Dataiku validation advantage:** Dataiku provides a flow-based process in which each transformation step produces an inspectable dataset, enabling teams to **filter, download, and validate** intermediate outputs more efficiently and reduce debugging time.
- ✚ **Monolithic architecture bottleneck:** The legacy design requires Power BI to execute compute-intensive transformations (joins, aggregations, complex calculations) while also rendering visual analytics. This combined workload constrains the BI engine and becomes a major bottleneck for stakeholders.

2.5 Strategic Transformation

The goal is to shift heavy transformation logic upstream and use Power BI primarily for high-performance visualization and analysis. The target-state approach includes:

- ✚ **Extract & Load (Dataiku):** The pipeline ingests and stages raw global oncology financial and sales data securely in Dataiku.
- ✚ **Transform & Clean (Dataiku):** Dataiku executes intensive joins, complex aggregations, and standardized business rules to generate an **analysis-ready final table**.
- ✚ **Visualize & Analyze (Power BI):** Power BI consumes the pre-processed dataset to improve refresh performance, simplify QC, reduce debugging effort, and deliver reliable insights for decision-making.

CHAPTER 3

METHODOLOGY

3.1 Overall Approach

The project follows an ELT-oriented methodology that shifts compute-intensive data preparation upstream to improve performance, auditability, and quality control. Raw global and regional Keytruda sales and financial datasets are ingested into Dataiku for secure staging, preprocessing, and transformation. Within Dataiku, the workflow applies standardized business rules, performs joins across relevant source tables, and executes aggregations to produce an **analysis-ready final dataset**. This transformed dataset then serves as the single source of truth for downstream reporting.

Power BI functions primarily as the visualization and analytics layer, consuming the curated table and implementing lightweight semantic measures where required. This separation of concerns reduces refresh latency, simplifies debugging through step-wise dataset inspection in Dataiku, and improves the reliability of quality checks prior to publication.

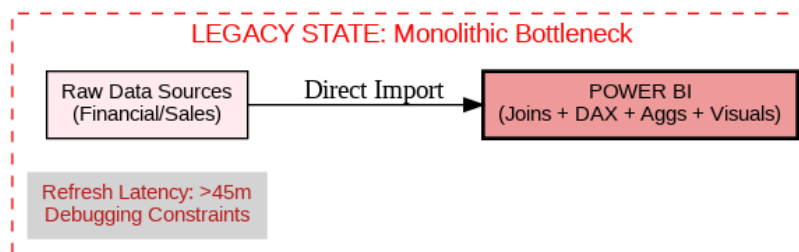


Figure 3.1: Legacy Architecture

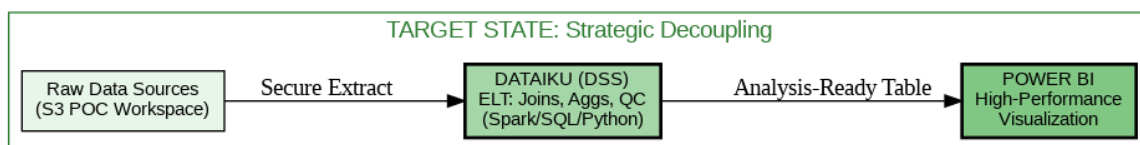


Figure 3.2: New Proposed Architecture

3.2 Tools & Tech Stack

Dataiku

- Secure data ingestion, staging, preprocessing, and transformation pipelines.
- Step-wise dataset inspection for QC and faster debugging.

Python

- Scripting for automation, custom data logic, and validation tasks.

Pandas (Python library)

- Data cleaning, manipulation, profiling, and reconciliation checks.

SQL

- Data extraction, joins, filtering, and business-rule implementation.
- Scripting for automation, custom data logic, and validation tasks.

Spark SQL

- High-performance SQL processing on large-scale datasets in distributed environments.

PySpark

- Distributed data transformations and aggregations for high-volume data processing.

Power BI

- Interactive dashboards, KPI reporting, and stakeholder-facing visual analytics.

Power Query

- Data connectivity and light transformations to support report readiness.

3.3 Timeline

Week	Phase	Description of Work
1-4	Phase 1: Requirement Analysis & System Discovery	Conduct a deep-dive review of existing Power BI reports and refresh behaviors to understand the Global Oncology financial transformation logic, KPIs, filters, and data dependencies.
5-6	Phase 2: Planning & Logic Migration	Map Power Query and DAX-based transformation logic to Dataiku recipes, defining the required joins, aggregations, and input-output specifications for the migrated design.
7-10	Phase 3: Data Migration	Relocate source datasets to the Amazon S3 POC workspace while configuring secure access controls and validating data completeness for downstream Dataiku processing.
11-15	Phase 4: Dataiku Pipeline Construction	Build an automated ETL pipeline in Dataiku by implementing transformations and joins to generate staged and final output tables with step-wise debugging capabilities.
16-17	Phase 5: QC Checks	Perform record-count checks and totals reconciliation against legacy outputs to ensure financial integrity and validate that KPI definitions are accurate.
18-19	Phase 6: Power BI Integration	Integrate optimized Dataiku final tables into Power BI datasets to reduce local transformation load.
20-22	Phase 7: UAT (User Acceptance Testing)	Facilitate stakeholder walkthroughs to validate dashboard usability and KPI correctness, incorporating feedback to obtain formal sign-off for the production release.
23	Phase 8: Launch	Deploy the final reporting pipeline and dashboards into production, ensuring scheduled refresh stability and delivering comprehensive release notes and handover documentation.

CHAPTER 4

DATA PREPARATION & ETL WITH DATAIKU

Dataiku is an enterprise data preparation and analytics platform that enables teams to build governed, reproducible pipelines for transforming raw data into analysis-ready datasets. It supports all: no-code, low-code development and code-driven engineering, allowing analysts and engineers to collaborate within a single environment. In this project, Dataiku functions as the primary upstream processing layer, where complex transformations are executed before the data is exposed to downstream BI consumption, thereby improving refresh performance, auditability, and quality control.

A core construct in Dataiku is the **Flow**, which provides a visual representation of the end-to-end data lineage from source inputs to curated outputs. The Flow is designed to make each transformation step explicit and traceable, so that intermediate datasets can be inspected and validated at every stage. This stepwise transparency is particularly valuable in high-stakes financial reporting, where stakeholders require confidence that the final metrics are derived through consistent logic and controlled transformations.

Transformations within the Flow are implemented through **Recipes**, which are modular processing steps that consume one or more datasets and generate new output datasets. Recipes can be configured using built-in visual components for standard data preparation tasks, or implemented as code recipes when custom logic, scalability, or performance optimization is required. This modular design supports maintainability, because business rules and transformation logic remain encapsulated in discrete, auditable steps rather than embedded across multiple reporting layers.

Dataiku supports both **ETL** and **ELT** paradigms, depending on where computation is best executed. In an ETL pattern, Dataiku extracts data from source systems, performs transformations within its execution environment, and loads curated outputs to a target. In an ELT pattern, data is first loaded into scalable storage or a data warehouse, and transformations are pushed down to the compute layer (for example, using warehouse SQL execution or Spark). In practice, this project adopts an ELT-oriented approach: raw data is staged in scalable platforms and Dataiku orchestrates

transformations close to the data, while still providing governance, lineage, and validation capabilities through the Flow.

4.1 Data Storage

Amazon S3 serves as a foundational storage layer for staging and exchanging datasets. In the project pipeline, Dataiku connects to S3 through a connection, enabling it to read raw files and write intermediate or curated outputs in standardized formats suitable for downstream processing. S3-based staging supports scalability and flexibility, particularly when data arrives from multiple sources and requires consistent partitioning, retention, and access control practices.

Amazon Redshift operates as a structured analytical warehouse that stores curated tables and supports SQL-based analytics at scale. Dataiku connects to Redshift using a dedicated database connection, which allows datasets to be imported as Dataiku-managed objects and enables transformations to be executed through SQL pushdown where appropriate. This integration supports efficient processing of large volumes of structured data, since compute-intensive joins and aggregations can be executed directly in the warehouse environment rather than transferring large datasets across systems. The ability to write outputs back to Redshift also supports standardized consumption patterns across reporting and analytics teams.

4.2 Language support

Within the Dataiku Flow, code-based processing is implemented through recipes such as **SQL**, **PySpark**, and **Python**. **SQL recipes** are used when transformations align naturally with relational operations such as filtering, joining, deduplication, and aggregation. **PySpark recipes** are used for distributed processing when datasets are large, transformations are computationally heavy, or when parallelization materially improves runtime. PySpark also enables more programmatic control over transformation logic while still operating within a scalable execution framework.

4.3 Flow

The Dataiku Flow serves as the backbone of the new reporting architecture. It shifts heavy transformation logic out of Power BI and into a governed data preparation environment.

It improves:

Performance, by reducing the computational burden on the BI layer.

Transparency, by exposing every transformation step visually.

Auditability, by preserving dataset lineage and reproducible outputs.

Quality control, by enabling validation of intermediate datasets.

Maintainability, by organizing logic into modular recipes instead of one monolithic report layer.

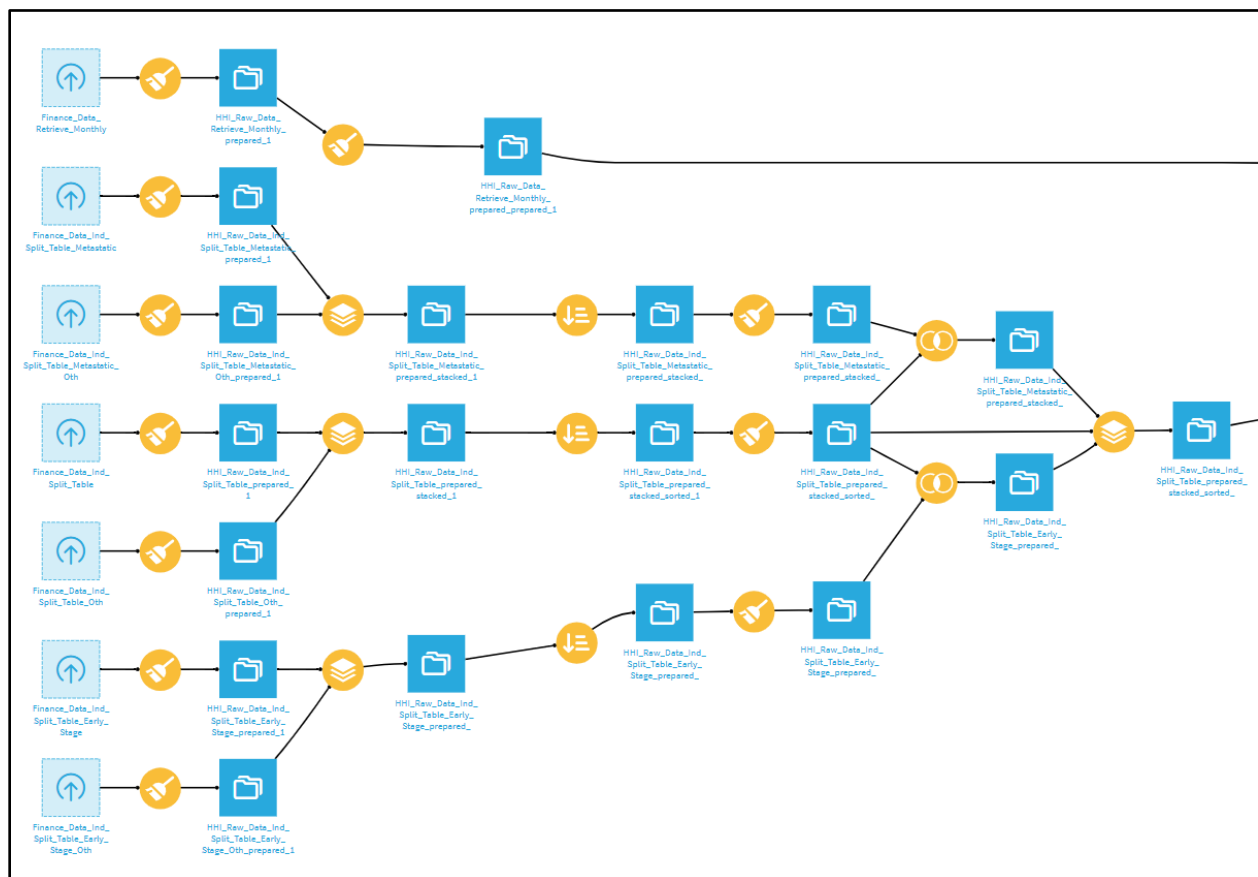


Figure 4.1: Dataiku Flow

The Dataiku Flow illustrated above represents the structured transformation pipeline through which multiple raw financial datasets are converted into a final, analysis-ready reporting table. The Flow provides a visual map of dataset lineage, showing how data moves from source inputs through a sequence of preparation and transformation steps before reaching the final curated output.

Components of the Flow:

Raw Input Datasets

- The blue nodes on the far left with the upload-style icon represent the initial raw datasets entering the Dataiku environment.
- These datasets are the source files that contain the unprocessed financial and sales data required for reporting.
- At this stage, the data includes inconsistencies such as non-standard column names, varying date formats, missing values, duplicate records.
- These raw inputs form the starting point of the ETL/ELT pipeline and are not yet suitable for direct analytical use.

Managed Dataset Nodes

- The blue folder-style nodes represent datasets stored or referenced within Dataiku.
- These nodes serve as controlled data objects that can be consumed by downstream recipes.
- Each dataset acts as a checkpoint in the pipeline, allowing the team to inspect, validate, and reuse the transformed output generated at a particular stage.
- This structure improves governance because every step produces a materialized dataset that can be independently reviewed.

Preparation Recipes

- The yellow circular icons with the brush-like symbol represent **Preparation Recipes**.
- Preparation recipes are used to clean and standardize individual datasets before they are combined with other sources.
- Typical actions performed in these recipes include:
 - Renaming columns to standardized business-friendly names.
 - Changing data types for dates, numeric fields, and dimensions.

- In this pipeline, join recipes integrate cleaned datasets with staging tables, split logic to produce intermediate outputs.
- The join step ensures that all relevant attributes and measures are brought together in a single dataset for further transformation.

Additional Transformation Recipes

- Some yellow recipe icons represent other intermediate transformation steps like:
 - Grouping and aggregation of records.
 - Appending or stacking similar datasets.
 - Filtering rows based on business rules.
 - Restructuring tables for reporting granularity.
 - Creating calculated fields or logic-driven subsets.

Intermediate Output Datasets

- After each recipe, a new blue dataset node is created to store the output of that transformation step.
- These intermediate datasets are highly valuable for validation and debugging.
- The team can open these tables, apply filters, download samples, and compare outputs against expected values during QC.
- This stepwise visibility is a major advantage over implementing the same logic entirely in Power BI, where intermediate transformed tables are not as easily accessible for validation.

Branching Structure in the Flow

- The Flow shows multiple branches progressing in parallel and later converging into common outputs.
- This branching structure indicates that different groups of source datasets are first processed independently before being integrated.
- Such a design is useful when different input streams represent distinct business components, such as:
 - Sales values
 - Financial splits
 - Time-based staging tables

- Market-level or regional mappings
- Processing these branches separately improves modularity, because logic changes in one branch can be validated without disrupting the entire pipeline.

Final Curated Dataset

- The blue dataset node on the far right represents the final analysis-ready output generated by the Dataiku Flow.
- This final table contains the cleaned, integrated, and transformed data required for downstream reporting in Power BI.
- By the time data reaches this stage, the major ETL/ELT tasks have already been completed:
 - Raw inputs are cleaned.
 - Structural inconsistencies are resolved.
 - Joins are completed.
 - Business logic is applied.

CHAPTER 5

DATA GOVERNANCE AND QUALITY TESTING

Data governance and quality testing form a critical part of the project because the transformed dataset is intended for high-stakes financial and business reporting. Once the final output table is generated in Dataiku, it is not directly accepted for downstream use without validation. Instead, the output undergoes a structured quality assurance process to confirm that the migrated transformation logic has been implemented correctly and that the resulting dataset is fully consistent with the legacy reporting output. This step is essential to ensure trust, traceability, and numerical reliability before the new pipeline is integrated into Power BI for stakeholder consumption.

A key component of this validation process is the comparison between the **final Dataiku output table** and the **final table generated through Power Query in the existing Power BI process**. The purpose of this comparison is to verify that all transformation steps previously handled in Power Query have been accurately mapped into the Dataiku pipeline. In other words, the migration is considered successful only if both tables are structurally and numerically equivalent, subject to agreed formatting or technical differences that do not affect business meaning. This comparison provides assurance that the new upstream transformation logic reproduces the same business output as the legacy implementation.

The quality testing process includes multiple levels of validation. At the structural level, the two outputs are checked to ensure that column names, field definitions, data types, and row-level granularity align correctly. At the content level, the datasets are compared for row counts, key combinations, and aggregated business values such as totals, subtotals, and KPI-related measures. Additional checks are performed by applying filters across dimensions such as country, time period, stage, and indication to confirm that the transformed data behaves consistently under different reporting scenarios. These checks help identify whether discrepancies arise from join conditions, filter logic, missing records, or differences in aggregation rules.

From a governance perspective, this comparison-based testing process strengthens confidence in the migrated pipeline by making validation explicit and reproducible. Rather than relying only on final dashboard outputs, the project validates the underlying transformed table itself, which is a more robust method for confirming transformation parity. Because Dataiku provides access to intermediate datasets at each stage of the Flow, we can investigate any mismatch in a controlled and transparent manner, trace the issue back to the relevant recipe, and correct the logic before proceeding further. This capability significantly improves debugging efficiency compared with a reporting environment in which intermediate transformation states are difficult to inspect.

Thus, data governance in this project extends beyond access control or storage management; it also includes transformation traceability, logic validation, and reproducible quality assurance. The goal is not only to generate a performant final dataset, but also to ensure that the dataset remains governed, explainable, and fully aligned with the legacy business logic on which stakeholders already rely. Through this validation framework, the project establishes that the Dataiku-based pipeline maintains both technical correctness and business continuity while supporting a more scalable reporting architecture.

CHAPTER 6

INTEGRATION OF DATAIKU OUTPUT INTO POWER BI

After the final output table is generated and validated in Dataiku, the next phase focuses on integrating this curated dataset into the Power BI reporting environment. This step is essential because it completes the transition from the legacy reporting architecture, in which extensive transformation logic is handled through Power Query, to the new architecture, in which most of the data preparation takes place in Dataiku. The objective of this phase is not only to replace the old source table, but also to ensure that the Power BI report continues to function correctly with respect to data model behavior, DAX calculations, and front-end visual outputs.

6.1 Replacement of the Legacy Table

The first activity in this phase involves replacing the existing legacy table in the Power BI data model with the final transformed table produced in Dataiku. The original table, which was previously made by joining multiple tables in Power Query and depended on multiple Power Query steps for cleaning, joining, and shaping the data, is disconnected from the active model. In its place, the Dataiku-generated final table is loaded into Power BI as the new reporting source.

This replacement requires careful attention to schema consistency. Column names, data types, key fields, and table granularity must remain aligned with the original model so that existing report components continue to function without major redesign. If the structure of the new table has been already made in Dataiku such that it doesn't differ much from the original source.

6.2 Removal of the Obsolete Power Query Logic

Once the new table is integrated, the outdated Power Query logic is removed from the report. The removal of these steps simplifies the report model and ensures that Power BI no longer acts as the primary transformation engine.

The key benefits of removing the legacy Power Query logic include:

- ✚ Reduced refresh complexity within Power BI.

- ✚ Improved report maintainability.
- ✚ Clear separation between data engineering and visualization layers.
- ✚ Lower dependency on in-report transformation steps.
- ✚ Better alignment with the new upstream ETL/ELT architecture.

6.3 Validation of DAX Measures

After the legacy table is replaced, the existing **DAX measures** are to be tested thoroughly to ensure that they continue to work correctly with the new Dataiku-based source table. Even though the underlying transformed dataset is intended to replicate the original output, any variation in schema, relationships, naming conventions, or filter context can affect DAX behavior.

The validation of DAX measures focuses on the following:

- ✚ Confirming that KPI calculations return the expected values
- ✚ Testing measures across different filters and slicer selections
- ✚ Verifying aggregations at different reporting levels, such as global, regional, and market views
- ✚ Checking that time-based calculations behave correctly
- ✚ Ensuring that ratio, percentage, and variance measures remain consistent with legacy outputs

This measure-level validation ensures that the semantic layer of the report continues to produce accurate business insights after the migration.

6.4 Front-End Report Validation

Following backend integration and DAX testing, validation is extended to the report front end.

Front-end validation includes:

- ✚ Checking that charts, tables, and KPI cards display accurate values.
- ✚ Ensuring that slicers and filters interact properly with visuals.
- ✚ Verifying drill-down and drill-through functionality where applicable.
- ✚ Reviewing page navigation and report responsiveness.
- ✚ Confirming that no visuals break due to changed source dependencies.

CHAPTER 7

LAUNCH AND CONCLUSION

After successful backend integration, DAX validation, and front-end testing, the optimized reporting solution proceeds to the launch stage. In this phase, the updated Power BI report is deployed into the intended business environment so that stakeholders can begin using the new reporting pipeline in practice. The launch marks the completion of the migration from the legacy Power Query-based transformation model to the new Dataiku-enabled architecture.

The launch process typically involves:

- ✚ Publishing the updated report with the new Dataiku-based source table.
- ✚ Verifying refresh schedules and confirming successful execution.
- ✚ Monitoring initial report's performance and stability.
- ✚ Ensuring appropriate user access and permissions.
- ✚ Communicating readiness to stakeholders and support teams.

7.1 Operational Benefits of the New Architecture

By moving heavy transformation logic out of Power BI and into Dataiku, the reporting architecture becomes more efficient, transparent, and maintainable. In the legacy setup, the final reporting table is derived from **seven source tables** undergoing **more than 210+ Power Query transformation steps** within Power BI. This created a substantial computational burden on the BI layer and contributed to prolonged refresh cycles that extend to **up to 1.5 hours**. With these transformation steps shifted upstream into Dataiku, the Power BI model becomes significantly lighter, which should lead to a considerable reduction in refresh time.

The benefits include:

- ✚ **Improved refresh performance**, as the removal of seven heavily transformed source tables and 210+ Power Query steps from Power BI should significantly reduce the overall refresh duration.

- ✚ **Better debugging and quality control (QC)**, because intermediate datasets in Dataiku can be inspected, filtered, and validated much more easily than transformation logic embedded in Power Query or DAX.
- ✚ **Enhanced auditability and lineage**, since recipe-level visibility in Dataiku makes the transformation path from raw source data to final output transparent and traceable.
- ✚ **Greater maintainability**, as business logic is organized into modular Dataiku recipes rather than distributed across numerous Power Query transformations.
- ✚ **Stronger scalability**, because Dataiku provides a more suitable environment for handling large and complex global financial datasets than a report-centric transformation model.

These improvements directly address the key limitations of the legacy reporting model and indicate that future quarterly refreshes will be faster, more reliable, and easier to validate.

7.2 Conclusion

From a broader perspective, the project highlights the value of adopting modern data engineering practices within enterprise reporting environments. When transformation logic is moved into a governed data pipeline and reporting tools are allowed to focus primarily on visualization and analysis, the resulting solution becomes more resilient, efficient, and easier to scale. This approach is especially relevant in high-volume financial reporting settings, where performance, accuracy, and traceability are all critical requirements.

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List of Abbreviations:

- ✚ **MSD:** Merck, Sharp & Dhome
- ✚ **BI:** Business Intelligence
- ✚ **DAX:** Data Analysis Expressions
- ✚ **PBI:** Power BI
- ✚ **ETL:** Extract, Transform, Load
- ✚ **PY:** Prior-Year
- ✚ **DHH:** Digital Human Health
- ✚ **HPV:** Human Papillomavirus
- ✚ **IC:** Insights Consumption
- ✚ **HCP:** HealthCare Professional
- ✚ **HCO:** HealthCare Organization
- ✚ **MR:** Medical Representative
- ✚ **KPIs:** Key Performance Indicators
- ✚ **POC:** Proof of Concept
- ✚ **QC:** Quality Check
- ✚ **UAT:** User Acceptance Testing
- ✚ **ELT:** Extract, Load, Transform